Berk Yilmaz

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Experience

Software Developer Dec 2024 – Present
Vretta (Systems team) *Toronto, ON*

- Led performance testing to keep fixed-date online exams stable for **195,000+ students** across **4 provinces**, using **k6** on **Kubernetes (AWS EKS)** and analyzing real traffic with **Athena** and **CloudWatch**.
- Lowered planned infrastructure cost for **2** provincial exam periods by an estimated **26–32%**, using load test data to halve the planned **AWS RDS** database and add **EC2** API servers where forecast to be needed.
- Derived a capacity rule of one API server per **1,000 students** a day from **k6** and **CloudWatch** data that matched 2025 production within **1%**; production then ran **12,800 students** a day on **15** servers with room to spare.
- Found **16%** of exam starts failed at **5**× production's login rate by simulating **8,000 concurrent students** in **k6**.
- Raised production traffic covered by load tests from **60% to 92%** on a province-wide program by modelling real usage from **AWS ALB** logs in **Athena**, and added a check that flags any live endpoint a test misses.
- Traced **2,781** production errors in a provincial exam's worst hour to a single application defect through **CloudWatch** log analysis.
- Protected student privacy by proving an existing data-masking script would have left **177,130** student IDs and birth dates exposed in an **Aurora MySQL** test copy of production data, then writing its replacement.
- Patched **5** authentication vulnerabilities in a **Node.js** exam platform API, shipping fixes to production for **2** clients to protect confidential student data.
- Built separate school, district and ministry admin access for an exam portal, using **11** role-specific **Feathers.js** endpoints, **32** hooks and **52** updated **Angular** components.
- Built out the production error-alerting pipeline to **2** more provinces on **AWS Lambda**, **Athena**, **Firehose** and **DynamoDB** with **GitLab CI**, keeping each province's data in its own AWS region.

Full-Stack Web Developer May 2024 – Dec 2024
Expense Trend (Contract) *Calgary, AB*

- Built personal finance APIs for community settings and per-bank account status (**Node.js**, **MSSQL**, **OpenAPI**).

Projects

Departure Brief | *TypeScript, React, Fastify, PostgreSQL* live demo | source

- Built and launched a free web app that combines **5** public aviation data sources into one preflight briefing for pilots and shows the original source behind every finding, backed by **899** automated tests.
- Used **Cloudflare Workers** and **KV** to serve live weather refreshed up to every **5** minutes, at no hosting cost.
- Wrote a **TypeScript** parser turning coded weather reports into plain text (<**0.3%** unparsed, **8,000** reports).
- Combined fixed rules with a **local LLM (Ollama)** to rank airspace notices by importance, requiring a quoted source for every claim; in a test on **31** real notices the system flagged every critical one at **$0** running cost.
- Caught that the project's AI quality test could not fail: rules settled **22 of 28** cases before the model ran, so a model giving one fixed answer would score **96%**; rebuilt it to grade the model only on what the rules leave open.

Hours to Go | *TypeScript, React, Fastify, PostgreSQL* live demo | source

- Built a web app that shows pilots how close they are to each of **9** Transport Canada licences and ratings, with every requirement cited to the regulation it comes from.
- Added accounts (**scrypt** password hashing), instant saving that reverts on failure, and mergeable guest logbooks.

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL
Frameworks: Node.js, Express, Feathers.js, Fastify, React, Angular, REST APIs, OpenAPI
Developer Tools: AWS (Lambda, EKS, EC2, RDS, S3, Athena, Firehose, DynamoDB, CloudWatch), Cloudflare Workers, Terraform, Kubernetes, Docker, GitLab CI/CD, k6
Data & AI: PostgreSQL, Aurora/MySQL, MSSQL, Redis, local LLMs, LLM evaluation, OCR
Certifications: Private Pilot Licence (Aeroplane), Transport Canada

Education

Wilfrid Laurier University Waterloo, ON
Honours Bachelor of Science in Computer Science